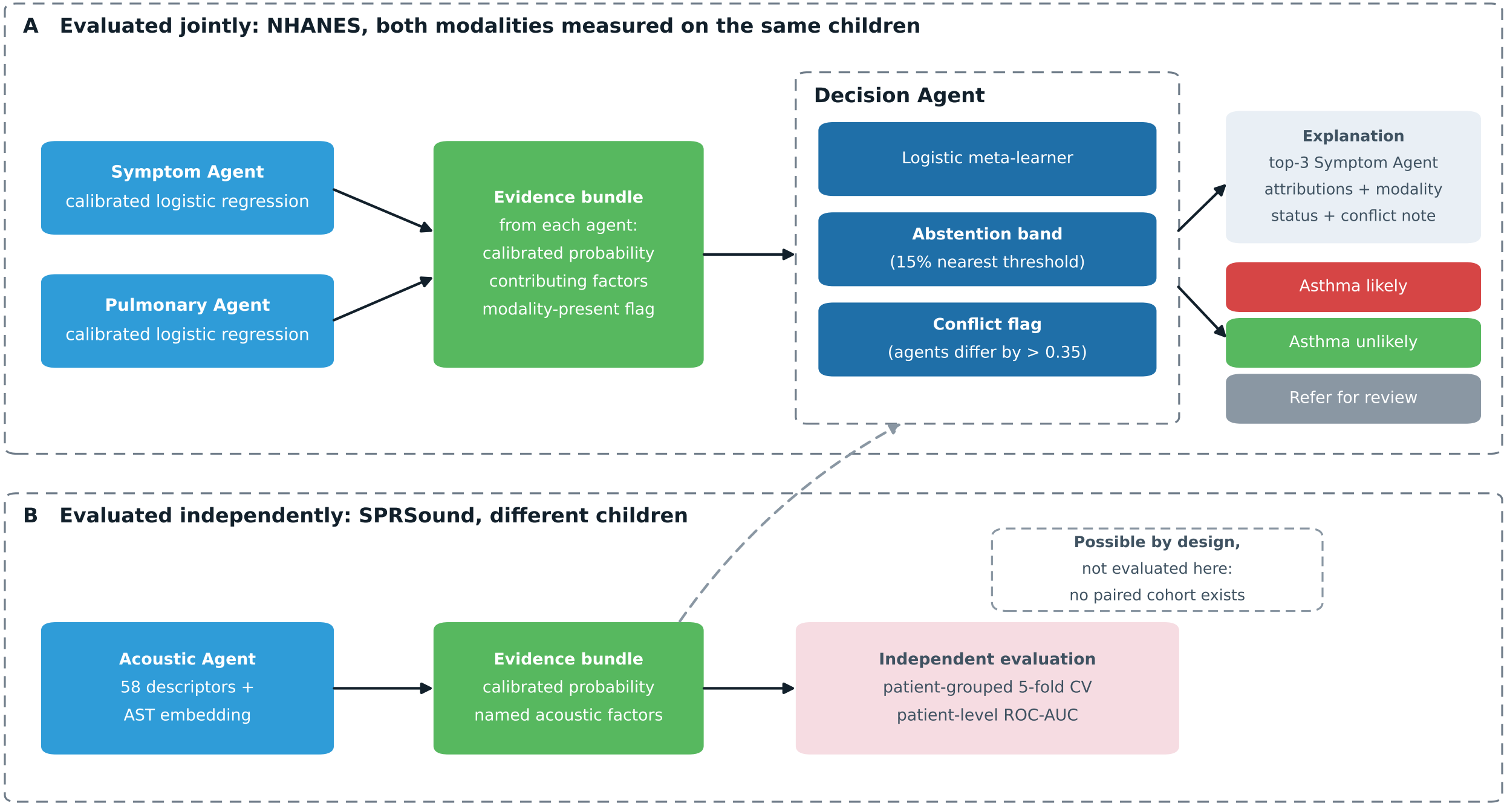
A Evaluated jointly: NHANES, both modalities measured on the same children
Symptom Agent
calibrated logistic regression
Pulmonary Agent
calibrated logistic regression
Evidence bundle
from each agent:
calibrated probability
contributing factors
modality-present flag
Decision Agent
Logistic meta-learner
Abstention band
(15% nearest threshold)
Conflict flag
(agents differ by > 0.35)
Explanation
top-3 Symptom Agent
attributions + modality
status + conflict note
Asthma likely
Asthma unlikely
Refer for review
B Evaluated independently: SPRSound, different children
Possible by design,
not evaluated here:
no paired cohort exists
Acoustic Agent
58 descriptors +
AST embedding
Evidence bundle
calibrated probability
named acoustic factors
Independent evaluation
patient-grouped 5-fold CV
patient-level ROC-AUC